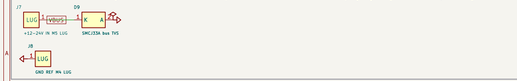
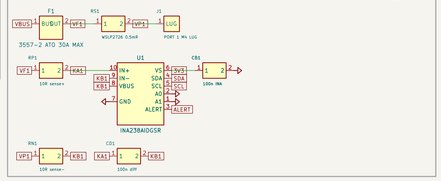
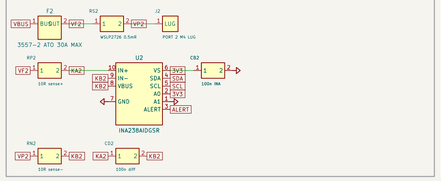
## 1. BUS INPUT: M5 lug 60A + SMCJ33A clamp; GND is REFERENCE only (D1)  [ADR-0001/0005]



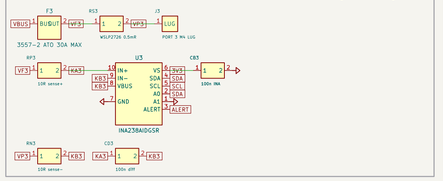

## 2.1 PORT 1: ATO fuse –> 0.5m shunt –> M4 stud; INA238 @0x40  [ADR-0003]



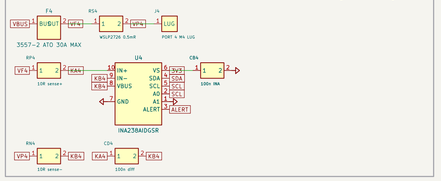

## 2.2 PORT 2: ATO fuse –> 0.5m shunt –> M4 stud; INA238 @0x41  [ADR-0003]




## 2.3 PORT 3: ATO fuse –> 0.5m shunt –> M4 stud; INA238 @0x42  [ADR-0003]




## 2.4 PORT 4: ATO fuse –> 0.5m shunt –> M4 stud; INA238 @0x43  [ADR-0003]




## 2.5 PORT 5: ATO fuse –> 0.5m shunt –> M4 stud; INA238 @0x44  [ADR-0003]




## 2.6 PORT 6: ATO fuse –> 0.5m shunt –> M4 stud; INA238 @0x45  [ADR-0003]




## 3. ELECTRONICS: 2A fuse –> reverse-block –> LMR16006X 60V buck –> 3.3V  [ADR-0001/0006]




## 4. ESP32-C3-WROOM-02: BLE 5.0 + native USB; straps per ADR-0004




## 5. LOG FLASH: W25Q64JV 8MB stats ring (dedicated, not app flash)  [ADR-0004]




## 6. USB-C: flash/debug + bench power (AMS1117 –> B5819W OR, ~3.0V)  [ADR-0006]




## 7. LEDS + DEBUG UART (J10 DNP)




12–24V/60A bus bar: 6x ATO-fused 30A ports, INA238+WSLP2726 sensing, ESP32-C3 BLE, W25Q64 l

Sheet: /
File: ble_bus_bar.kicad_sch

**Title: ble-bus-bar**

Size: A1 | Date: 2026-07-19 | **Rev: v1.0**
KiCad E.D.A. 10.0.4 | Id: 1/1